

DESi A/B Evidence — LongMemEval-s Full Sweep

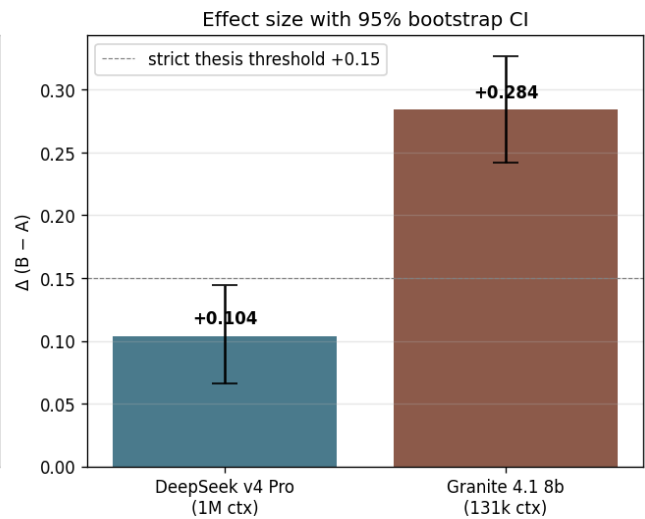
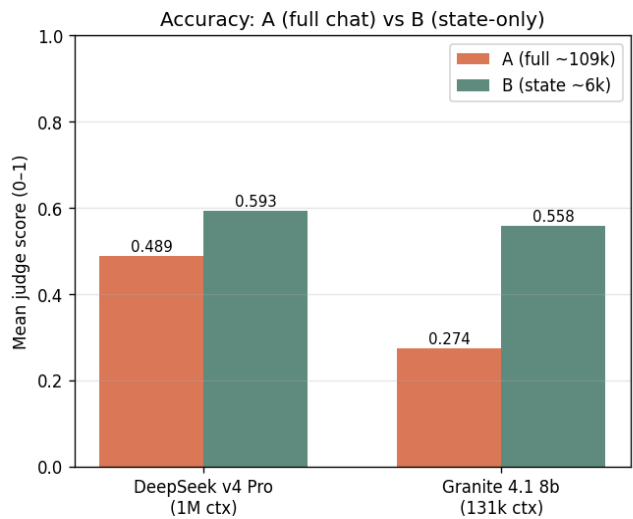
Independent peer-published benchmark · N=500 · DeepSeek v4 Pro + Granite 4.1 8b · OpenAI GPT-4o as judge · cost \$27.91 total

Headline result (overall, 95% bootstrap CI)

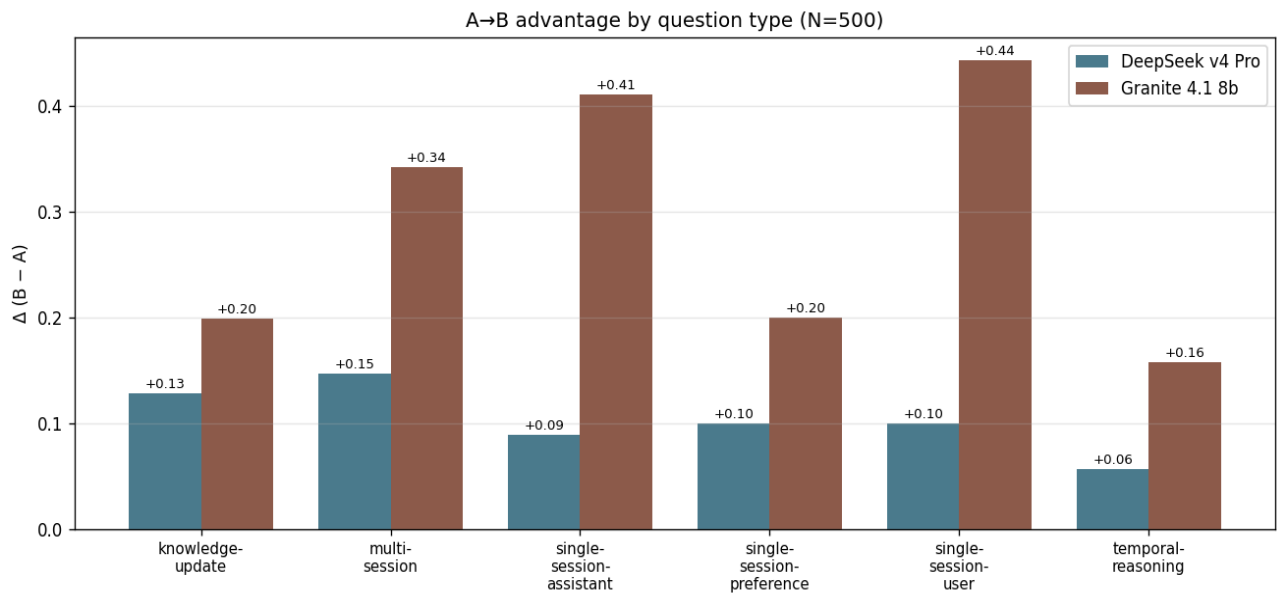
Model	A	B	Δ (B–A)	95% CI	B>A%	B=A%	B<A%	A fail
deepseek-v4-pro	0.489	0.593	+0.104	[+0.066, +0.144]	23	66	11	0
granite-4.1-8b	0.274	0.558	+0.284	[+0.241, +0.326]	44	47	8	52

Pre-registered hypothesis status

Hypothesis	DeepSeek	Granite
Weak (B > A statistically significant)	✓ CONFIRMED	✓ CONFIRMED
Strict ($\Delta \geq +0.15$)	✗ REFUTED (upper CI +0.144)	✓ CONFIRMED (+0.284)
No falsification criteria triggered	—	—

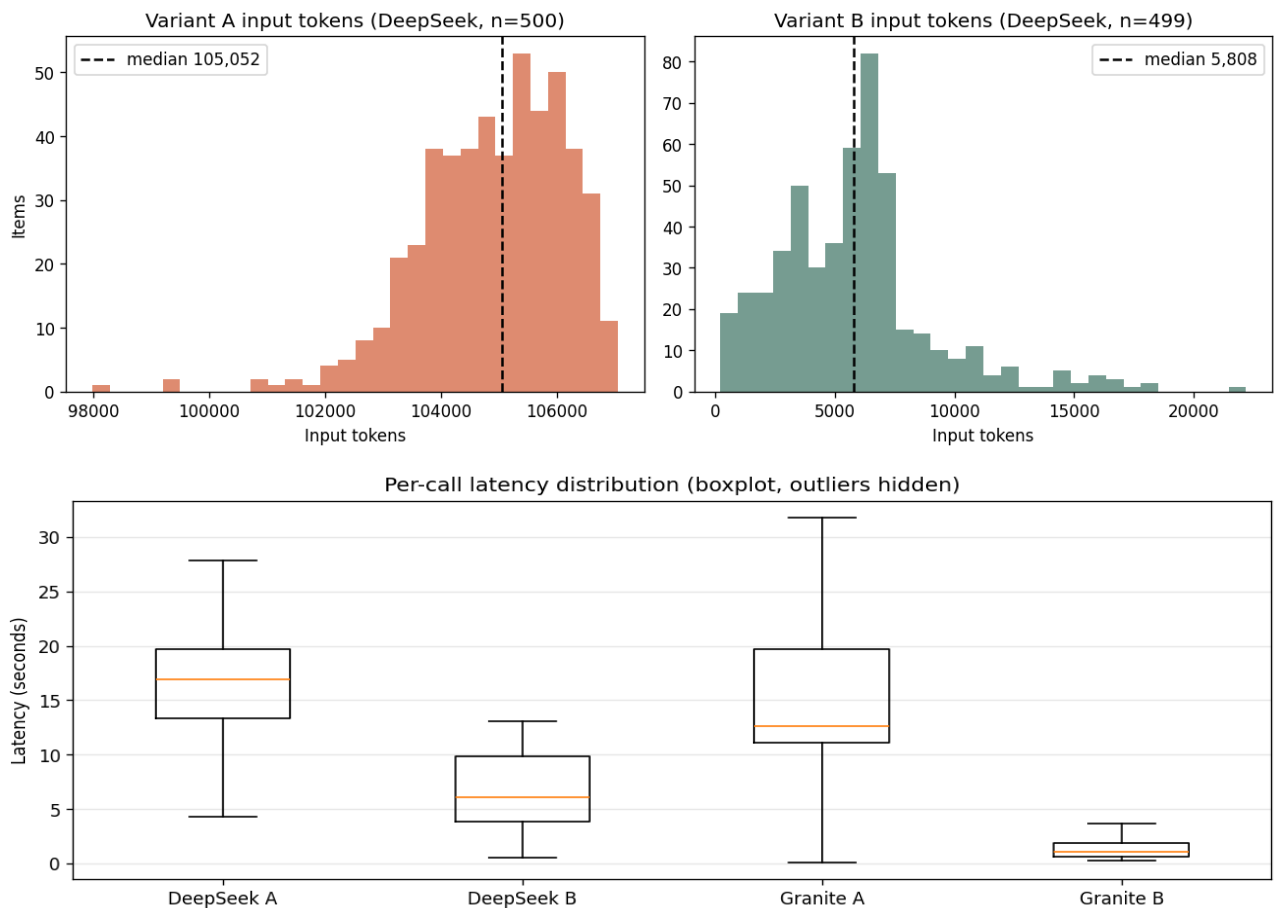


Per question type (N=500)



qtype	n	DS A	DS B	ΔDS	GR A	GR B	ΔGR	GR fail
knowledge-update	78	0.615	0.744	+0.128	0.449	0.647	+0.199	5/78
multi-session	133	0.451	0.598	+0.147	0.150	0.492	+0.342	16/133
single-session-assistant	56	0.848	0.938	+0.089	0.438	0.848	+0.411	5/56
single-session-preference	30	0.267	0.367	+0.100	0.333	0.533	+0.200	4/30
single-session-user	70	0.807	0.907	+0.100	0.371	0.814	+0.443	6/70
temporal-reasoning	133	0.184	0.241	+0.056	0.162	0.320	+0.158	16/133

Tokens, cost, latency



model	var	mean_in	mean_out	sum_in	sum_out	mean_lat_s	cost \$
deepseek-v4-pro	A	104,908	370	52,454,124	185,008	16.278	\$22.72
deepseek-v4-pro	B	5,792	304	2,890,313	151,628	6.734	\$1.37
granite-4.1-8b	A	110,387	134	49,453,174	59,982	14.053	\$2.48
granite-4.1-8b	B	6,125	113	3,062,271	56,639	1.921	\$0.16
judge (GPT-4o)	—	—	—	—	—	—	\$1.18
TOTAL							\$27.91

Refusal rates ('I don't know')

Model	A refusals %	B refusals %
deepseek-v4-pro	6.8	14.0
granite-4.1-8b	4.4	2.8

B refuses more, not less. When the curated subset lacks info, the model correctly says 'I don't know' instead of hallucinating from noise. A in the same situation tries to answer and is wrong.

Per-item data (all 500 items)

Columns: question_id, qtype, A_tok = input tokens A, B_tok = input tokens B, ds_A/ds_B = DeepSeek scores, gr_A/gr_B = Granite scores. Failures (e.g. Granite at 131k limit) are recorded as score 0.0.

question_id	qtype	A_tok	B_tok	ds_A	ds_B	gr_A	gr_B
01493427	knowledge-update	104805	6513	0.0	1.0	0.0	1.0
031748ae	knowledge-update	106492	5305	0.0	1.0	0.0	0.5
031748ae_abs	knowledge-update	106944	5301	0.0	1.0	0.5	0.0
06db6396	knowledge-update	105835	7082	1.0	0.5	1.0	1.0
07741c44	knowledge-update	105434	5807	0.5	0.5	0.0	0.5
07741c45	knowledge-update	103273	5807	0.0	0.0	1.0	0.0
08e075c7	knowledge-update	105266	6223	1.0	1.0	1.0	1.0
0977f2af	knowledge-update	104844	5854	1.0	1.0	1.0	0.0
0ddfec37	knowledge-update	103418	5601	0.5	1.0	0.0	1.0
0ddfec37_abs	knowledge-update	104828	5604	1.0	1.0	0.5	1.0
0e4e4c46	knowledge-update	104777	6676	1.0	1.0	1.0	0.5
0f05491a	knowledge-update	104234	4077	1.0	0.5	0.5	1.0
10e09553	knowledge-update	104006	5442	0.5	0.5	0.0	0.0
184da446	knowledge-update	103638	8209	0.0	1.0	1.0	1.0
18bc8abd	knowledge-update	102541	7555	1.0	1.0	1.0	1.0
1cea1afa	knowledge-update	106404	6692	1.0	0.5	0.0	1.0
2133c1b5	knowledge-update	105449	5557	0.0	1.0	0.0	0.5
2133c1b5_abs	knowledge-update	102508	5561	0.0	1.0	0.5	1.0
22d2cb42	knowledge-update	104181	5212	1.0	0.5	1.0	0.5
2698e78f	knowledge-update	106759	6007	0.0	0.0	0.0	0.5
2698e78f_abs	knowledge-update	105954	6008	1.0	1.0	0.5	1.0
26bdc477	knowledge-update	105928	6639	1.0	0.5	0.0	1.0
3ba21379	knowledge-update	103856	9110	1.0	1.0	0.0	1.0
41698283	knowledge-update	104495	7078	0.0	1.0	1.0	0.0
42ec0761	knowledge-update	104293	5873	0.5	0.5	0.0	0.5
45dc21b6	knowledge-update	104450	6326	1.0	1.0	0.0	1.0
4b24c848	knowledge-update	105308	4700	1.0	1.0	1.0	1.0
4d6b87c8	knowledge-update	102751	6157	1.0	1.0	1.0	1.0
50635ada	knowledge-update	106122	5808	0.0	0.5	0.0	0.0
5831f84d	knowledge-update	103733	6292	1.0	1.0	1.0	1.0
59524333	knowledge-update	103768	5195	0.0	0.5	0.0	1.0
5a4f22c0	knowledge-update	106436	5879	1.0	1.0	0.0	1.0
5c40ec5b	knowledge-update	104463	4819	1.0	1.0	1.0	1.0
603deb26	knowledge-update	106715	6713	1.0	1.0	1.0	1.0
6071bd76	knowledge-update	105241	4896	1.0	1.0	0.0	1.0
618f13b2	knowledge-update	105754	5965	1.0	1.0	0.0	1.0
69fee5aa	knowledge-update	104908	6116	0.0	0.0	0.0	0.0

question_id	qtype	A_tok	B_tok	ds_A	ds_B	gr_A	gr_B
6a1eabeb	knowledge-update	104092	7495	1.0	0.0	1.0	0.0
6a27ffc2	knowledge-update	103993	8606	1.0	1.0	1.0	1.0
6aeb4375	knowledge-update	102592	5143	1.0	0.5	1.0	0.0
6aeb4375_abs	knowledge-update	104747	5147	0.0	1.0	0.5	0.5
71315a70	knowledge-update	105399	7502	1.0	1.0	1.0	1.0
72e3ee87	knowledge-update	104132	6288	1.0	1.0	1.0	1.0
7401057b	knowledge-update	103551	6848	0.0	1.0	0.0	0.0
7a87bd0c	knowledge-update	105411	6244	1.0	1.0	0.0	1.0
7e974930	knowledge-update	103409	4516	1.0	1.0	1.0	1.0
830ce83f	knowledge-update	103963	5224	0.0	0.0	0.0	0.0
852ce960	knowledge-update	104107	5890	1.0	0.0	0.0	0.0
89941a93	knowledge-update	106195	7107	0.5	1.0	1.0	1.0
89941a94	knowledge-update	105145	7123	0.0	0.0	0.5	0.5
8fb83627	knowledge-update	104187	5174	1.0	0.0	1.0	1.0
945e3d21	knowledge-update	106482	6223	0.0	1.0	0.5	0.5
9bbe84a2	knowledge-update	104361	6141	0.5	0.5	0.0	1.0
9ea5eabc	knowledge-update	104640	7301	0.0	1.0	0.5	0.0
a1eacc2a	knowledge-update	105021	6981	1.0	1.0	1.0	1.0
a2f3aa27	knowledge-update	104482	5369	0.5	0.0	0.0	0.0
affe2881	knowledge-update	105201	6305	1.0	1.0	0.0	1.0
b01defab	knowledge-update	106171	5208	1.0	0.5	0.0	0.5
b6019101	knowledge-update	105987	5373	1.0	1.0	1.0	1.0
ba61f0b9	knowledge-update	106369	6175	0.0	0.0	1.0	0.5
c4ea545c	knowledge-update	104776	5616	0.0	1.0	0.5	0.5
c6853660	knowledge-update	105462	5470	1.0	1.0	1.0	1.0
c7dc5443	knowledge-update	105987	6873	1.0	1.0	0.0	0.0
cc5ded98	knowledge-update	103925	6194	1.0	0.5	0.0	1.0
ce6d2d27	knowledge-update	105456	5602	0.5	1.0	1.0	1.0
cf22b7bf	knowledge-update	104265	7559	1.0	1.0	1.0	0.5
d7c942c3	knowledge-update	106600	5100	0.5	0.5	0.0	0.5
dad224aa	knowledge-update	105755	5736	0.0	0.0	0.0	0.5
db467c8c	knowledge-update	105037	7435	1.0	1.0	1.0	0.5
dfde3500	knowledge-update	104606	4893	1.0	1.0	0.0	1.0
e493bb7c	knowledge-update	106804	6622	0.5	1.0	0.0	0.5
e61a7584	knowledge-update	105157	5685	0.0	1.0	0.0	1.0
e66b632c	knowledge-update	105823	6498	0.0	1.0	0.0	0.0
eace081b	knowledge-update	106746	6559	1.0	1.0	0.0	1.0
ed4ddc30	knowledge-update	105052	7720	0.0	1.0	0.0	0.0
f685340e	knowledge-update	103375	4873	1.0	0.0	0.0	0.5
f685340e_abs	knowledge-update	106256	4870	0.0	1.0	1.0	1.0

question_id	qtype	A_tok	B_tok	ds_A	ds_B	gr_A	gr_B
f9e8c073	knowledge-update	106273	7212	1.0	0.5	0.0	0.0
00ca467f	multi-session	105291	10087	1.0	0.0	0.0	1.0
0100672e	multi-session	106764	6064	1.0	1.0	0.0	0.5
078150f1	multi-session	104163	6868	0.0	0.5	1.0	1.0
099778bb	multi-session	106681	5974	0.0	1.0	0.0	0.0
09ba9854	multi-session	102926	5082	0.0	0.0	0.0	0.5
09ba9854_abs	multi-session	105494	5086	0.0	0.0	0.5	0.0
0a995998	multi-session	106007	8309	0.0	0.0	0.0	0.0
0ea62687	multi-session	103904	6127	1.0	0.5	0.0	1.0
10d9b85a	multi-session	105870	6972	1.0	1.0	0.0	1.0
1192316e	multi-session	105446	6863	1.0	1.0	0.0	0.0
129d1232	multi-session	103862	10103	0.0	0.0	0.0	0.0
157a136e	multi-session	105396	7101	0.5	1.0	0.0	0.0
1a8a66a6	multi-session	105264	10671	1.0	0.0	0.0	0.5
1c549ce4	multi-session	104837	6165	1.0	1.0	0.0	1.0
1f2b8d4f	multi-session	103921	6673	1.0	1.0	0.0	0.5
21d02d0d	multi-session	106650	5770	0.0	0.0	0.5	0.0
2311e44b	multi-session	104805	4948	0.5	1.0	0.0	0.5
2311e44b_abs	multi-session	105682	4951	1.0	1.0	0.0	1.0
2318644b	multi-session	102488	6966	1.0	1.0	0.0	0.5
27016adc	multi-session	106173	8435	1.0	1.0	0.0	1.0
2788b940	multi-session	103342	12471	1.0	0.0	0.0	0.0
28dc39ac	multi-session	105049	18086	0.0	0.0	0.0	0.0
2b8f3739	multi-session	106739	10249	0.0	1.0	0.0	0.0
2ce6a0f2	multi-session	106150	11116	0.0	0.0	0.0	0.0
2e6d26dc	multi-session	103721	10533	0.0	0.0	0.0	0.0
36b9f61e	multi-session	104898	9829	1.0	1.0	0.0	0.0
37f165cf	multi-session	104591	6064	0.0	0.0	0.0	0.0
3a704032	multi-session	106667	10142	0.0	0.0	0.0	0.0
3c1045c8	multi-session	100975	7875	0.5	1.0	0.0	1.0
3fdac837	multi-session	103146	3773	0.0	0.0	0.0	0.0
3fe836c9	multi-session	105592	7073	1.0	1.0	1.0	1.0
46a3abf7	multi-session	105572	11036	0.0	0.0	0.0	1.0
4adc0475	multi-session	106527	7888	1.0	1.0	0.0	0.0
4bc144e2	multi-session	105439	3916	1.0	1.0	1.0	1.0
4f54b7c9	multi-session	99464	6504	0.0	1.0	0.5	0.5
5025383b	multi-session	103196	6417	1.0	1.0	0.5	0.5
51c32626	multi-session	105207	5687	0.0	0.0	0.5	0.5
55241a1f	multi-session	105585	6665	1.0	1.0	0.0	1.0
5a7937c8	multi-session	103341	9465	0.5	0.0	0.0	0.5

question_id	qtype	A_tok	B_tok	ds_A	ds_B	gr_A	gr_B
60036106	multi-session	103956	7094	0.0	1.0	1.0	1.0
60159905	multi-session	106541	6478	0.0	0.5	0.0	0.0
60472f9c	multi-session	97990	10652	1.0	1.0	0.0	1.0
60bf93ed	multi-session	104628	err	1.0	0.0	0.0	1.0
60bf93ed_abs	multi-session	104167	6086	0.0	1.0	1.0	1.0
61f8c8f8	multi-session	106116	5970	1.0	1.0	0.0	0.5
6456829e	multi-session	104924	6486	1.0	1.0	0.0	0.0
6456829e_abs	multi-session	103815	6491	0.0	1.0	0.5	1.0
67e0d0f2	multi-session	105675	6987	0.0	1.0	0.0	0.0
681a1674	multi-session	106230	6779	0.0	1.0	0.0	1.0
6c49646a	multi-session	105869	5981	0.5	1.0	0.0	0.0
6cb6f249	multi-session	103650	6760	1.0	1.0	0.0	1.0
6d550036	multi-session	99493	14140	0.0	0.0	0.0	0.0
7024f17c	multi-session	104482	9340	0.5	0.0	0.0	1.0
720133ac	multi-session	106793	4602	1.0	1.0	0.0	1.0
73d42213	multi-session	103808	4917	0.0	0.0	0.0	0.0
7405e8b1	multi-session	106155	5457	0.0	0.0	0.0	0.5
77eafa52	multi-session	105405	4184	1.0	1.0	0.0	1.0
80ec1f4f	multi-session	104763	7054	0.0	1.0	0.0	1.0
80ec1f4f_abs	multi-session	103829	7057	0.0	1.0	0.0	1.0
81507db6	multi-session	106553	15213	1.0	1.0	0.0	0.5
85fa3a3f	multi-session	105939	3984	1.0	1.0	0.0	1.0
87f22b4a	multi-session	105012	6654	1.0	1.0	0.0	1.0
88432d0a	multi-session	105237	12270	0.0	0.0	0.0	0.0
88432d0a_abs	multi-session	105885	12280	0.5	1.0	0.5	0.5
8979f9ec	multi-session	105491	6307	1.0	1.0	0.0	1.0
8cf4d046	multi-session	104267	8651	1.0	1.0	0.5	1.0
8e91e7d9	multi-session	105114	6727	0.0	1.0	0.0	1.0
91b15a6e	multi-session	105053	6610	1.0	0.5	0.0	0.0
92a0aa75	multi-session	104761	7060	0.0	0.0	0.0	0.0
9aaed6a3	multi-session	104961	2498	0.0	1.0	0.0	1.0
9d25d4e0	multi-session	106068	9737	0.0	0.0	0.0	0.0
9ee3ecd6	multi-session	104725	5262	0.0	0.0	0.0	0.0
a08a253f	multi-session	102129	5723	0.0	0.0	0.0	0.0
a11281a2	multi-session	106806	8177	0.0	0.0	0.0	1.0
a1cc6108	multi-session	104086	5926	1.0	1.0	0.0	1.0
a3332713	multi-session	104366	3597	1.0	1.0	0.0	1.0
a346bb18	multi-session	106510	6651	1.0	1.0	0.0	1.0
a4996e51	multi-session	106665	6428	1.0	1.0	0.5	1.0
a96c20ee	multi-session	104177	6116	1.0	1.0	1.0	1.0

question_id	qtype	A_tok	B_tok	ds_A	ds_B	gr_A	gr_B
a96c20ee_abs	multi-session	106167	6123	0.0	0.0	1.0	0.0
a9f6b44c	multi-session	105538	6786	0.0	0.0	0.5	0.5
aae3761f	multi-session	103669	9600	0.5	1.0	0.0	0.5
b3c15d39	multi-session	101675	5761	1.0	0.5	1.0	1.0
b5ef892d	multi-session	105156	10172	1.0	0.0	0.0	0.0
ba358f49	multi-session	104895	6365	0.0	1.0	0.0	1.0
ba358f49_abs	multi-session	104366	6366	0.5	1.0	0.5	1.0
bb7c3b45	multi-session	103739	5925	0.5	1.0	0.0	0.5
bc149d6b	multi-session	102969	5609	0.0	0.5	0.0	1.0
bf659f65	multi-session	102925	8286	0.0	0.0	0.0	0.0
c18a7dc8	multi-session	105215	7208	1.0	1.0	0.0	1.0
c2ac3c61	multi-session	105607	7541	1.0	1.0	0.0	0.0
c4a1ceb8	multi-session	104801	12283	0.0	0.0	0.0	0.0
cc06de0d	multi-session	105692	4823	1.0	1.0	0.0	0.0
d23cf73b	multi-session	104774	14146	0.0	0.0	0.5	1.0
d3ab962e	multi-session	104291	5961	0.5	1.0	0.0	1.0
d6062bb9	multi-session	105102	6499	0.0	1.0	0.0	1.0
d682f1a2	multi-session	104400	9432	1.0	1.0	0.0	0.5
d851d5ba	multi-session	104515	12387	0.0	1.0	0.0	0.0
d905b33f	multi-session	105716	4606	0.0	1.0	1.0	0.0
dd2973ad	multi-session	105377	5600	1.0	0.5	0.0	0.0
e25c3b8d	multi-session	105987	5339	0.0	1.0	1.0	1.0
e3038f8c	multi-session	103995	13357	0.0	0.0	0.0	0.0
e56a43b9	multi-session	105179	5165	1.0	1.0	1.0	0.5
e5ba910e_abs	multi-session	102819	5712	0.0	0.5	0.5	0.5
e6041065	multi-session	104651	5741	1.0	1.0	0.0	1.0
e831120c	multi-session	104195	6015	0.0	1.0	0.0	1.0
edced276	multi-session	104711	5893	1.0	1.0	0.0	0.0
edced276_abs	multi-session	103988	5895	0.5	1.0	0.5	0.5
eeda8a6d	multi-session	106154	6415	1.0	1.0	0.0	1.0
eeda8a6d_abs	multi-session	104371	6417	0.0	0.5	0.5	0.5
ef66a6e5	multi-session	103677	6320	1.0	0.0	0.0	1.0
ef9cf60a	multi-session	106306	5850	0.5	1.0	0.0	0.0
efc3f7c2	multi-session	106057	3243	1.0	0.5	0.0	1.0
f0e564bc	multi-session	102759	7373	1.0	1.0	0.0	0.0
f35224e0	multi-session	105936	5431	1.0	1.0	0.0	0.0
gpt4_15e38248	multi-session	104414	10111	0.0	1.0	0.0	0.0
gpt4_194be4b3	multi-session	104570	16354	1.0	0.0	0.0	0.5
gpt4_2ba83207	multi-session	106652	11608	0.0	0.0	0.0	0.0
gpt4_2f8be40d	multi-session	104114	10920	0.0	0.0	0.0	0.0

question_id	qtype	A_tok	B_tok	ds_A	ds_B	gr_A	gr_B
gpt4_2f91af09	multi-session	105642	11075	0.5	0.0	0.0	0.0
gpt4_31ff4165	multi-session	103086	16226	0.0	1.0	0.0	0.0
gpt4_372c3eed	multi-session	105318	11525	0.0	0.0	0.0	0.0
gpt4_372c3eed_abs	multi-session	105338	11531	0.0	0.0	0.5	0.0
gpt4_5501fe77	multi-session	105668	10638	0.0	0.0	0.0	0.0
gpt4_59c863d7	multi-session	105160	15925	0.0	0.0	0.0	0.5
gpt4_731e37d7	multi-session	105407	11975	0.0	0.0	0.0	0.0
gpt4_7fce9456	multi-session	105433	17277	0.0	0.0	0.0	0.0
gpt4_a56e767c	multi-session	105062	10694	0.0	1.0	0.0	0.5
gpt4_ab202e7f	multi-session	105544	16118	0.0	0.0	0.5	0.5
gpt4_d12ceb0e	multi-session	102697	9949	1.0	1.0	0.0	1.0
gpt4_d84a3211	multi-session	104732	11505	0.0	0.5	0.0	0.5
gpt4_e05b82a6	multi-session	105745	11060	0.0	0.0	0.0	0.0
gpt4_f2262a51	multi-session	104152	9694	0.0	1.0	0.5	1.0
0e5e2d1a	single-session-assista	105814	728	1.0	1.0	0.0	1.0
1568498a	single-session-assista	104062	1989	1.0	1.0	0.0	0.5
16c90bf4	single-session-assista	103481	1536	1.0	0.5	0.5	0.5
18dcd5a5	single-session-assista	105762	975	1.0	1.0	1.0	1.0
1903aded	single-session-assista	104527	231	1.0	1.0	0.0	1.0
1b9b7252	single-session-assista	104139	1085	1.0	1.0	1.0	0.5
1d4da289	single-session-assista	106451	777	1.0	1.0	0.0	1.0
1de5cff2	single-session-assista	105977	1253	1.0	1.0	1.0	1.0
28bcfaac	single-session-assista	106419	770	1.0	1.0	0.0	1.0
2bf43736	single-session-assista	103798	2129	1.0	1.0	0.5	1.0
3249768e	single-session-assista	106412	724	0.5	1.0	0.0	1.0
352ab8bd	single-session-assista	105686	2075	1.0	1.0	0.5	1.0
3e321797	single-session-assista	106791	709	1.0	1.0	1.0	1.0
41275add	single-session-assista	105892	354	1.0	1.0	0.0	1.0
4388e9dd	single-session-assista	104100	1294	1.0	1.0	0.5	1.0
488d3006	single-session-assista	106359	1393	0.5	1.0	0.5	1.0
4baee567	single-session-assista	103314	1692	0.5	1.0	1.0	0.5
4c36cccf	single-session-assista	105782	765	1.0	1.0	0.5	1.0
51b23612	single-session-assista	103228	1513	0.5	1.0	0.0	0.5
561fabcd	single-session-assista	105526	2126	0.5	1.0	0.5	0.0
5809eb10	single-session-assista	104876	2086	1.0	1.0	1.0	1.0
58470ed2	single-session-assista	105363	1192	1.0	1.0	1.0	0.5
6222b6eb	single-session-assista	106628	2113	1.0	1.0	0.0	1.0
65240037	single-session-assista	105246	2011	0.5	1.0	1.0	1.0
6ae235be	single-session-assista	104393	1639	1.0	1.0	0.5	1.0
70b3e69b	single-session-assista	105910	2336	1.0	1.0	0.0	1.0

question_id	qtype	A_tok	B_tok	ds_A	ds_B	gr_A	gr_B
7161e7e2	single-session-assista	104492	2182	0.0	1.0	0.0	1.0
71a3fd6b	single-session-assista	105000	568	1.0	1.0	0.5	1.0
778164c6	single-session-assista	102026	1127	0.5	0.0	0.5	1.0
7a8d0b71	single-session-assista	105458	2240	1.0	1.0	0.5	1.0
7e00a6cb	single-session-assista	104846	828	1.0	0.5	0.0	0.5
8464fc84	single-session-assista	104564	1235	1.0	1.0	0.0	0.5
8752c811	single-session-assista	104073	1789	0.0	1.0	0.0	0.0
89527b6b	single-session-assista	105834	755	1.0	1.0	1.0	1.0
8aef76bc	single-session-assista	105962	905	1.0	1.0	1.0	1.0
8b9d4367	single-session-assista	104696	989	1.0	1.0	1.0	1.0
8cf51dda	single-session-assista	105458	465	1.0	1.0	1.0	0.5
a40e080f	single-session-assista	105662	961	1.0	1.0	0.0	1.0
ac031881	single-session-assista	103883	2053	0.5	0.5	0.5	0.5
b759caee	single-session-assista	105363	666	1.0	1.0	0.0	1.0
c4f10528	single-session-assista	105674	1755	1.0	0.5	0.5	1.0
c7cf7dfd	single-session-assista	103617	1948	1.0	1.0	1.0	1.0
c8f1aeed	single-session-assista	103864	1064	1.0	1.0	1.0	1.0
cc539528	single-session-assista	104559	1235	0.5	0.5	0.5	0.5
ceb54acb	single-session-assista	105360	333	1.0	1.0	1.0	1.0
d596882b	single-session-assista	105530	1233	1.0	1.0	0.0	1.0
dc439ea3	single-session-assista	105505	1137	1.0	1.0	0.0	1.0
e3fc4d6e	single-session-assista	104153	1046	0.5	1.0	1.0	1.0
e48988bc	single-session-assista	105447	779	0.5	1.0	0.0	1.0
e8a79c70	single-session-assista	106547	1077	1.0	1.0	0.5	1.0
e9327a54	single-session-assista	104392	2098	1.0	1.0	0.0	1.0
e982271f	single-session-assista	106057	730	1.0	1.0	0.0	0.0
eaca4986	single-session-assista	104197	1322	0.0	1.0	0.0	1.0
f523d9fe	single-session-assista	106547	1251	1.0	1.0	0.0	1.0
fca762bc	single-session-assista	105565	694	1.0	1.0	0.5	1.0
fea54f57	single-session-assista	104488	1644	1.0	1.0	0.5	1.0
06878be2	single-session-prefere	105330	4560	0.0	0.0	0.0	0.0
06f04340	single-session-prefere	103906	4045	0.5	0.0	0.5	1.0
07b6f563	single-session-prefere	104892	4136	0.0	0.0	0.5	1.0
09d032c9	single-session-prefere	104026	6171	0.0	0.0	0.5	0.5
0a34ad58	single-session-prefere	104872	5316	0.5	0.0	0.0	0.5
0edc2aef	single-session-prefere	106519	3950	0.0	0.0	0.0	1.0
195a1a1b	single-session-prefere	104856	4061	0.5	1.0	0.5	0.5
1a1907b4	single-session-prefere	106259	3859	0.0	1.0	0.0	0.5
1c0ddc50	single-session-prefere	106394	3876	0.0	1.0	0.5	0.5
1d4e3b97	single-session-prefere	105215	3437	1.0	1.0	0.5	0.5

question_id	qtype	A_tok	B_tok	ds_A	ds_B	gr_A	gr_B
1da05512	single-session-prefere	106518	4412	0.5	0.0	0.5	0.5
32260d93	single-session-prefere	104009	3615	0.0	0.5	0.0	0.0
35a27287	single-session-prefere	102320	4248	0.0	0.0	0.0	0.0
38146c39	single-session-prefere	106275	4572	0.0	0.0	0.5	0.5
505af2f5	single-session-prefere	105820	3773	1.0	1.0	0.5	1.0
54026fce	single-session-prefere	106649	3003	0.5	1.0	0.5	1.0
57f827a0	single-session-prefere	106069	3610	0.0	0.0	0.5	0.0
6b7dfb22	single-session-prefere	106204	4698	0.0	0.0	0.5	0.5
75832dbd	single-session-prefere	103335	3902	0.0	0.0	0.0	1.0
75f70248	single-session-prefere	105517	3770	0.0	1.0	0.0	0.5
8a2466db	single-session-prefere	105576	5458	1.0	0.5	0.5	0.5
95228167	single-session-prefere	104794	3455	0.0	0.0	0.5	0.5
a89d7624	single-session-prefere	104917	3048	1.0	0.5	0.0	0.5
afdc33df	single-session-prefere	106589	4775	0.0	0.0	0.0	0.5
b0479f84	single-session-prefere	104152	3789	1.0	1.0	0.5	1.0
b6025781	single-session-prefere	107062	2812	0.0	0.0	0.5	0.5
caf03d32	single-session-prefere	105586	4706	0.5	0.0	0.5	0.5
d24813b1	single-session-prefere	105009	4353	0.0	1.0	0.5	0.0
d6233ab6	single-session-prefere	105192	2792	0.0	0.5	0.5	0.5
fca70973	single-session-prefere	104532	3445	0.0	0.0	0.5	0.5
001be529	single-session-user	105623	3292	1.0	1.0	0.0	0.5
0862e8bf	single-session-user	100794	2881	1.0	1.0	1.0	1.0
0862e8bf_abs	single-session-user	105082	2883	0.5	0.0	1.0	1.0
118b2229	single-session-user	103564	2926	1.0	1.0	0.0	1.0
15745da0	single-session-user	106335	3922	1.0	1.0	0.0	1.0
15745da0_abs	single-session-user	103557	3924	0.5	1.0	1.0	0.5
19b5f2b3	single-session-user	103189	3591	1.0	1.0	0.5	1.0
19b5f2b3_abs	single-session-user	104626	3593	0.0	1.0	1.0	1.0
1e043500	single-session-user	105587	2883	1.0	1.0	0.0	1.0
1faac195	single-session-user	106212	3317	0.5	1.0	0.0	1.0
21436231	single-session-user	104266	3794	1.0	1.0	0.0	1.0
25e5aa4f	single-session-user	105707	3952	1.0	1.0	0.0	1.0
29f2956b	single-session-user	102343	3986	1.0	1.0	0.0	1.0
29f2956b_abs	single-session-user	105788	3988	1.0	1.0	0.0	1.0
311778f1	single-session-user	106053	3534	1.0	1.0	0.0	1.0
36580ce8	single-session-user	103196	3426	1.0	1.0	0.5	1.0
37d43f65	single-session-user	105145	3378	1.0	1.0	1.0	1.0
3b6f954b	single-session-user	103621	2901	1.0	1.0	1.0	1.0
3d86fd0a	single-session-user	105805	1806	1.0	1.0	0.0	1.0
3f1e9474	single-session-user	104473	4034	0.0	1.0	0.5	1.0

question_id	qtype	A_tok	B_tok	ds_A	ds_B	gr_A	gr_B
4100d0a0	single-session-user	104609	2826	1.0	1.0	0.5	1.0
4fd1909e	single-session-user	106386	1857	0.5	0.5	1.0	1.0
51a45a95	single-session-user	105392	2395	0.0	0.0	0.0	0.5
545bd2b5	single-session-user	103395	3694	1.0	1.0	0.0	1.0
577d4d32	single-session-user	105036	2852	1.0	1.0	0.5	1.0
58bf7951	single-session-user	105546	2142	1.0	1.0	1.0	1.0
58ef2f1c	single-session-user	103661	2743	1.0	1.0	0.5	0.0
5d3d2817	single-session-user	105261	3460	1.0	1.0	0.0	1.0
60d45044	single-session-user	106023	3262	1.0	1.0	0.0	0.5
66f24dbb	single-session-user	103834	1421	0.5	0.5	0.0	0.5
6ade9755	single-session-user	104157	2044	0.5	1.0	0.5	0.5
6b168ec8	single-session-user	106441	3575	1.0	0.5	0.0	0.5
6f9b354f	single-session-user	106648	3190	1.0	1.0	0.0	1.0
726462e0	single-session-user	105364	3387	1.0	1.0	1.0	1.0
7527f7e2	single-session-user	105272	3581	1.0	1.0	1.0	0.5
75499fd8	single-session-user	104866	3073	1.0	1.0	0.0	0.5
76d63226	single-session-user	104101	3032	1.0	1.0	0.0	0.5
853b0a1d	single-session-user	103983	2593	1.0	1.0	0.0	1.0
8550ddae	single-session-user	106197	3596	1.0	1.0	0.0	0.5
86b68151	single-session-user	104015	2650	1.0	1.0	0.0	1.0
86f00804	single-session-user	102189	3225	0.0	1.0	0.0	1.0
8a137a7f	single-session-user	104486	3397	0.0	1.0	0.0	1.0
8e9d538c	single-session-user	105261	3594	1.0	1.0	0.0	1.0
8ebdbe50	single-session-user	106469	3909	0.5	1.0	1.0	1.0
94f70d80	single-session-user	104738	3125	1.0	0.5	1.0	1.0
95bcc1c8	single-session-user	106073	3663	1.0	1.0	1.0	1.0
a06e4cfe	single-session-user	106396	3387	1.0	0.5	0.0	0.5
a82c026e	single-session-user	106629	2571	1.0	1.0	1.0	1.0
ad7109d1	single-session-user	102394	3302	0.5	1.0	0.0	0.5
af8d2e46	single-session-user	103050	2846	1.0	1.0	0.0	0.5
b320f3f8	single-session-user	105213	4008	1.0	1.0	0.5	1.0
b86304ba	single-session-user	103219	2450	0.0	1.0	0.0	0.5
bc8a6e93	single-session-user	104289	3658	1.0	1.0	1.0	0.5
bc8a6e93_abs	single-session-user	106058	3660	1.0	1.0	1.0	1.0
c14c00dd	single-session-user	105291	3164	1.0	1.0	0.0	0.5
c19f7a0b	single-session-user	104958	2982	1.0	1.0	0.0	1.0
c5e8278d	single-session-user	105792	3351	0.5	1.0	1.0	1.0
c8c3f81d	single-session-user	106803	2678	0.5	1.0	0.0	0.5
c960da58	single-session-user	106131	3610	1.0	1.0	0.0	1.0
caf9ead2	single-session-user	105155	1720	1.0	1.0	0.0	1.0

question_id	qtype	A_tok	B_tok	ds_A	ds_B	gr_A	gr_B
ccb36322	single-session-user	105399	2526	0.0	1.0	0.5	0.5
d52b4f67	single-session-user	104822	2224	1.0	1.0	1.0	1.0
dccbc061	single-session-user	103760	4535	1.0	1.0	0.5	0.5
e01b8e2f	single-session-user	105967	3885	1.0	0.5	0.0	0.5
e47becba	single-session-user	104951	3809	1.0	1.0	1.0	1.0
ec81a493	single-session-user	106330	3065	0.0	0.5	0.0	0.5
f4f1d8a4	single-session-user	104990	2443	0.5	0.5	1.0	1.0
f4f1d8a4_abs	single-session-user	106374	2444	1.0	1.0	1.0	0.0
f8c5f88b	single-session-user	104596	3000	1.0	0.5	0.0	1.0
faba32e5	single-session-user	106115	3482	1.0	1.0	0.0	1.0
08f4fc43	temporal-reasoning	105384	6411	0.0	0.0	0.0	1.0
0bb5a684	temporal-reasoning	105965	7361	0.5	1.0	0.0	1.0
0bc8ad92	temporal-reasoning	104094	8466	0.0	0.0	0.0	0.0
0bc8ad93	temporal-reasoning	104462	8837	0.0	0.0	0.5	0.0
0db4c65d	temporal-reasoning	103494	7176	0.0	0.0	0.0	0.0
2a1811e2	temporal-reasoning	106574	6170	0.0	0.0	0.0	1.0
2c63a862	temporal-reasoning	106096	5683	1.0	1.0	0.0	0.5
2ebe6c90	temporal-reasoning	103106	6052	0.0	0.0	0.0	0.0
2ebe6c92	temporal-reasoning	104859	6042	0.0	0.0	1.0	1.0
370a8ff4	temporal-reasoning	103126	6510	0.0	0.0	0.0	0.0
4dfccbf7	temporal-reasoning	105640	6372	0.0	0.0	0.0	0.0
4dfccbf8	temporal-reasoning	105581	6357	0.0	0.0	0.0	0.0
5e1b23de	temporal-reasoning	104253	3603	0.0	0.5	0.0	0.0
6613b389	temporal-reasoning	105821	9044	1.0	0.0	1.0	1.0
6e984301	temporal-reasoning	106047	8338	0.0	0.0	0.0	0.0
6e984302	temporal-reasoning	103541	8332	0.0	0.5	0.0	0.0
71017276	temporal-reasoning	104154	3947	0.0	0.0	0.0	0.0
71017277	temporal-reasoning	106153	3939	0.0	0.0	0.0	0.0
8077ef71	temporal-reasoning	104596	1617	0.0	0.0	0.0	0.0
8c18457d	temporal-reasoning	104814	4622	0.5	0.0	0.0	1.0
982b5123	temporal-reasoning	106795	6058	0.0	0.0	0.0	0.5
982b5123_abs	temporal-reasoning	106524	6058	1.0	0.5	0.5	1.0
993da5e2	temporal-reasoning	103897	5589	0.0	0.0	0.0	1.0
9a707b81	temporal-reasoning	106063	7061	0.0	0.0	0.0	0.0
9a707b82	temporal-reasoning	105634	7055	0.0	0.5	0.0	0.5
a3045048	temporal-reasoning	104679	4957	0.0	1.0	1.0	0.0
a3838d2b	temporal-reasoning	103506	17838	0.0	0.0	0.0	0.0
af082822	temporal-reasoning	106134	2994	0.0	0.0	0.0	0.0
b29f3365	temporal-reasoning	106006	6375	0.0	0.0	0.0	0.5
b46e15ed	temporal-reasoning	106158	14379	0.0	0.0	0.0	0.0

question_id	qtype	A_tok	B_tok	ds_A	ds_B	gr_A	gr_B
b46e15ee	temporal-reasoning	103837	14377	0.0	0.0	0.0	0.5
b9cfe692	temporal-reasoning	105893	8261	1.0	1.0	0.0	1.0
bbf86515	temporal-reasoning	104368	6155	1.0	1.0	1.0	1.0
bcbe585f	temporal-reasoning	104782	2412	0.0	0.0	0.0	0.0
c8090214	temporal-reasoning	101128	6254	0.0	1.0	1.0	1.0
c8090214_abs	temporal-reasoning	104410	6255	0.0	1.0	0.5	1.0
c9f37c46	temporal-reasoning	104980	6793	0.0	0.0	0.0	0.5
cc6d1ec1	temporal-reasoning	104201	6522	0.0	0.0	0.0	0.5
d01c6aa8	temporal-reasoning	105273	7551	0.0	0.0	0.0	0.0
dcfa8644	temporal-reasoning	103054	5668	1.0	1.0	0.0	1.0
e4e14d04	temporal-reasoning	105938	6133	0.0	0.0	0.0	0.0
eac54adc	temporal-reasoning	106663	7491	0.0	0.0	0.0	0.0
eac54add	temporal-reasoning	103678	7495	0.0	0.0	0.0	0.0
f0853d11	temporal-reasoning	103443	7069	0.0	0.0	1.0	1.0
gpt4_0a05b494	temporal-reasoning	104347	6979	0.0	0.0	0.0	0.0
gpt4_0b2f1d21	temporal-reasoning	104340	6864	0.0	0.0	0.0	0.0
gpt4_18c2b244	temporal-reasoning	106867	8401	0.0	0.0	0.0	0.5
gpt4_1916e0ea	temporal-reasoning	106239	6042	0.0	0.0	0.0	0.0
gpt4_1a1dc16d	temporal-reasoning	106720	5974	1.0	1.0	0.0	1.0
gpt4_1d4ab0c9	temporal-reasoning	104317	6962	0.0	0.0	1.0	0.0
gpt4_1d80365e	temporal-reasoning	106050	6915	0.0	0.0	0.0	0.0
gpt4_1e4a8aeb	temporal-reasoning	106325	6107	0.0	0.0	0.0	0.0
gpt4_1e4a8aec	temporal-reasoning	105177	6604	0.0	0.0	0.0	0.0
gpt4_213fd887	temporal-reasoning	103331	7807	1.0	0.0	0.0	0.0
gpt4_21adecb5	temporal-reasoning	106220	8891	0.0	0.0	0.0	0.0
gpt4_2312f94c	temporal-reasoning	104472	5936	0.0	0.5	0.0	1.0
gpt4_2487a7cb	temporal-reasoning	105671	7055	0.0	0.0	0.0	1.0
gpt4_2655b836	temporal-reasoning	105732	8888	0.5	1.0	0.0	0.5
gpt4_2c50253f	temporal-reasoning	105530	6759	1.0	0.5	0.0	0.0
gpt4_2d58bcd6	temporal-reasoning	104227	7467	0.5	1.0	0.5	1.0
gpt4_2f56ae70	temporal-reasoning	104028	8466	0.0	1.0	0.0	1.0
gpt4_2f584639	temporal-reasoning	103265	5024	0.0	0.0	0.0	0.0
gpt4_385a5000	temporal-reasoning	103422	4695	0.5	0.5	1.0	0.0
gpt4_45189cb4	temporal-reasoning	103112	7521	0.0	0.0	0.0	0.0
gpt4_468eb063	temporal-reasoning	105314	3683	0.0	0.0	0.0	0.0
gpt4_468eb064	temporal-reasoning	103289	3685	0.0	0.0	0.0	0.0
gpt4_483dd43c	temporal-reasoning	106460	6441	0.0	0.0	0.5	0.0
gpt4_4929293a	temporal-reasoning	105899	6248	0.0	0.0	0.0	0.0
gpt4_4929293b	temporal-reasoning	105683	6196	0.0	0.0	0.0	0.5
gpt4_4cd9eba1	temporal-reasoning	106413	7525	1.0	1.0	0.0	0.5

question_id	qtype	A_tok	B_tok	ds_A	ds_B	gr_A	gr_B
gpt4_4edbafa2	temporal-reasoning	104880	6819	0.5	1.0	1.0	1.0
gpt4_4ef30696	temporal-reasoning	105515	6896	0.0	0.0	0.0	0.0
gpt4_4fc4f797	temporal-reasoning	105965	7031	0.0	0.0	0.0	0.0
gpt4_5438fa52	temporal-reasoning	103661	7989	0.0	0.0	1.0	0.0
gpt4_59149c77	temporal-reasoning	103523	7866	0.0	0.0	0.0	0.0
gpt4_59149c78	temporal-reasoning	105435	7855	0.0	0.0	0.0	0.0
gpt4_5dcc0aab	temporal-reasoning	105426	13694	1.0	0.5	1.0	1.0
gpt4_61e13b3c	temporal-reasoning	105581	7167	0.0	0.0	0.0	0.0
gpt4_65aabe59	temporal-reasoning	102963	7151	0.0	0.0	0.0	1.0
gpt4_68e94287	temporal-reasoning	104828	5763	0.5	0.0	1.0	1.0
gpt4_68e94288	temporal-reasoning	104832	6266	0.0	0.5	0.0	0.5
gpt4_6dc9b45b	temporal-reasoning	102673	4564	0.0	0.0	0.0	0.0
gpt4_6ed717ea	temporal-reasoning	103855	6778	0.0	0.0	0.0	0.0
gpt4_70e84552	temporal-reasoning	104786	7130	0.0	0.0	0.0	0.0
gpt4_70e84552_abs	temporal-reasoning	106168	7133	0.0	0.0	0.5	0.5
gpt4_74aed68e	temporal-reasoning	106016	6745	0.0	0.0	0.0	0.0
gpt4_76048e76	temporal-reasoning	104491	5642	1.0	1.0	0.0	0.0
gpt4_78cf46a3	temporal-reasoning	104671	5235	0.0	0.0	1.0	1.0
gpt4_7a0daae1	temporal-reasoning	105663	7120	0.0	0.0	0.0	0.0
gpt4_7abb270c	temporal-reasoning	101544	16105	0.0	0.0	0.0	0.0
gpt4_7bc6cf22	temporal-reasoning	105698	4240	0.0	0.0	0.0	0.0
gpt4_7ca326fa	temporal-reasoning	103792	7949	0.0	0.0	0.0	0.0
gpt4_7ddcf75f	temporal-reasoning	105052	3505	0.0	0.0	0.0	0.0
gpt4_7de946e7	temporal-reasoning	105278	5563	1.0	1.0	1.0	1.0
gpt4_7f6b06db	temporal-reasoning	103837	8964	0.0	0.0	0.0	1.0
gpt4_8279ba02	temporal-reasoning	103683	3509	0.0	0.0	0.0	0.0
gpt4_8279ba03	temporal-reasoning	105979	3602	0.0	0.5	0.0	0.5
gpt4_85da3956	temporal-reasoning	103810	2328	1.0	1.0	0.5	1.0
gpt4_88806d6e	temporal-reasoning	106788	5298	0.0	1.0	0.0	0.0
gpt4_8c8961ae	temporal-reasoning	103374	6850	1.0	0.0	0.5	0.0
gpt4_8e165409	temporal-reasoning	103813	6375	0.0	0.0	0.0	0.0
gpt4_93159ced	temporal-reasoning	103455	6783	0.0	1.0	0.0	1.0
gpt4_93159ced_abs	temporal-reasoning	101421	6786	0.0	1.0	0.5	0.5
gpt4_93f6379c	temporal-reasoning	104246	11157	1.0	1.0	0.0	1.0
gpt4_98f46fc6	temporal-reasoning	105336	6190	0.0	0.0	0.0	1.0
gpt4_9a159967	temporal-reasoning	106440	7927	0.0	0.0	0.0	0.0
gpt4_a1b77f9c	temporal-reasoning	106137	22178	0.0	0.0	0.0	0.0
gpt4_a2d1d1f6	temporal-reasoning	106242	3280	0.0	0.0	0.0	0.0
gpt4_af6db32f	temporal-reasoning	106233	2886	0.0	0.0	0.0	0.0
gpt4_b0863698	temporal-reasoning	106061	2956	0.0	0.0	0.0	0.0

question_id	qtype	A_tok	B_tok	ds_A	ds_B	gr_A	gr_B
gpt4_b4a80587	temporal-reasoning	106004	7066	0.0	1.0	0.0	1.0
gpt4_b5700ca0	temporal-reasoning	104830	2903	1.0	0.5	0.0	0.5
gpt4_b5700ca9	temporal-reasoning	103868	2911	0.0	0.0	0.0	0.0
gpt4_c27434e8	temporal-reasoning	106026	6791	0.0	0.0	0.0	0.0
gpt4_c27434e8_abs	temporal-reasoning	105497	6796	1.0	0.5	0.0	0.5
gpt4_cd90e484	temporal-reasoning	105123	6176	0.0	0.0	0.0	0.0
gpt4_d31cdae3	temporal-reasoning	101980	5662	0.0	1.0	1.0	0.0
gpt4_d6585ce8	temporal-reasoning	103218	16454	0.0	0.0	0.0	0.5
gpt4_d6585ce9	temporal-reasoning	103747	16683	0.0	0.0	0.0	0.0
gpt4_d9af6064	temporal-reasoning	104011	6588	1.0	1.0	0.0	0.0
gpt4_e061b84f	temporal-reasoning	103449	9632	0.0	0.0	0.0	1.0
gpt4_e061b84g	temporal-reasoning	104061	9627	0.0	1.0	0.0	0.0
gpt4_e072b769	temporal-reasoning	105627	2938	0.0	0.0	0.0	0.0
gpt4_e414231e	temporal-reasoning	106120	6563	0.0	0.0	0.0	0.0
gpt4_e414231f	temporal-reasoning	106398	6588	0.0	0.0	0.0	0.0
gpt4_ec93e27f	temporal-reasoning	105989	5104	0.5	0.0	0.0	1.0
gpt4_f420262c	temporal-reasoning	104877	14437	0.0	0.0	0.0	0.0
gpt4_f420262d	temporal-reasoning	105706	15175	0.0	0.0	0.5	0.0
gpt4_f49edff3	temporal-reasoning	104433	7413	0.0	0.0	0.5	0.0
gpt4_fa19884c	temporal-reasoning	102651	6335	0.0	0.0	0.0	0.0
gpt4_fa19884d	temporal-reasoning	105408	7031	0.0	0.5	0.0	0.0
gpt4_fe651585	temporal-reasoning	103836	6610	0.5	0.5	0.0	0.0
gpt4_fe651585_abs	temporal-reasoning	104830	6614	1.0	0.0	1.0	0.5

Honest caveats and what this study does NOT show

- B = LongMemEval's curated *answer_session_ids* subset (~6k tokens); NOT a structured DESi-state. Tests the same IDEA (relevant subset replaces full history) but not the specific DESi mechanism (Claims/Decisions/Constraints).
- GPT-4o single-judge introduces systematic bias; our judge is stricter than the paper's (our Sonnet 4.5 A on 15 items = 0.167 vs paper's Claude 3.5 Sonnet ~0.50). Δ direction is robust but absolute numbers shift with judge choice.
- Sonnet 4.5 and GPT-5 are NOT in this 500-item run, only indicative from the prior 15-item sample.
- Granite A hit the 131k context limit on 41/500 items (counted as score 0; this is the user-experienced failure, not an artifact correction).
- LongMemEval question distribution is a constructed mix from ShareGPT/UltraChat sources; generalization to other dialog types is not guaranteed.
- Run was interrupted 2× by container OOM/recycle; resume worked cleanly, all 500 items completed.
- API keys (DeepSeek + OpenRouter) shared in chat log; rotate after this study. Never written to any file in the repo (grep-verified).

What is robust across this study

- **Token reduction $A \rightarrow B = 94.5\%$** (mean and median, both models, $n=500$). Property of the variant.
- **$B > A$ in ALL 6 question-types on BOTH models.**
- **95% CIs exclude 0 on both models.** The weak hypothesis is statistically established.
- **B reduces hallucination by refusing more often when info is missing.** Across 500 items, B's refusal rate is 1.5–2× higher than A's — and these are correct refusals (gold-answer was not in curated subset).
- **Latency: 2.4× faster on DeepSeek, 7.3× faster on Granite for B.**
- **Cost: B is 16× cheaper than A on DeepSeek, 15× cheaper on Granite.**